```
> restart;
  with(student) :
  with(linalg) :
```

```
> 
  # ==================================================================
  #       4
  #    1
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```

```
> 

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  #    4 . 1
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```

```
> 
  eq := diff(u(x, t), t) - a^2 * diff(u(x, t), x$2) = 0;
  0 < x, x < L, 0 < t;
```

$$eq := \frac{\partial}{\partial t}\, u(x, t) - a^2 \left(\frac{\partial^2}{\partial x^2}\, u(x, t)\right) = 0$$

$$0 < x, x < L, 0 < t \tag{1}$$

```
> 
  init_c := u(x, 0) = phi(x);
  bound_c := u(0, t) = 0, u(L, t) = 0;
  phi := x    c*x*(L-x)/L^2;
```

$$init\_c := u(x, 0) = \phi(x)$$

$$bound\_c := u(0, t) = 0, u(L, t) = 0$$

$$\phi := x \quad \frac{c \cdot x \cdot (L - x)}{L^2} \tag{2}$$

```
> 
  # ==================================================================
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  _F1 := '_F1';
  _F2 := '_F2';

  eq1 := diff(_F1(x), x$2) + lambda*_F1(x) = 0;
```

$$\_F1 := \_F1$$

$$\_F2 := \_F2$$

$$eq1 := \frac{d^2}{dx^2}\,\_F1(x) + \lambda\,\_F1(x) = 0 \tag{3}$$

>

```
solF0 := dsolve(eq1, _F1(x));
_F1 := unapply(rhs(solF0), x);
```

$$solF0 := \_F1(x) = c_1 \sin\left(\sqrt{\lambda}\, x\right) + c_2 \cos\left(\sqrt{\lambda}\, x\right)$$

$$\_F1 := x \mapsto c_1 \cdot \sin\left(\sqrt{\lambda} \cdot x\right) + c_2 \cdot \cos\left(\sqrt{\lambda} \cdot x\right) \tag{4}$$

>

```
e1 := _F1(0) = 0;
e2 := _F1(L) = 0;
sist := {e1, e2};
```

$$e1 := c_2 = 0$$

$$e2 := c_1 \sin\left(\sqrt{\lambda}\, L\right) + c_2 \cos\left(\sqrt{\lambda}\, L\right) = 0$$

$$sist := \left\{c_2 = 0, c_1 \sin\left(\sqrt{\lambda}\, L\right) + c_2 \cos\left(\sqrt{\lambda}\, L\right) = 0\right\} \tag{5}$$

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```
with(linalg) :

A := genmatrix(sist, {c__1, c__2}) :
detA := convert(det(A), trig) :
detA;
```

$$-\sin\left(\sqrt{\lambda}\, L\right) \tag{6}$$

>

```
_EnvAllSolutions := true :
solLam := solve(detA = 0, lambda) :
solLam;
```

$$\frac{\pi^2\, \_Z1{\sim}^2}{L^2} \tag{7}$$

>

```
ev := k -> Pi^2*k^2/L^2 :
ev(k);
```

$$\frac{\pi^2 k^2}{L^2} \tag{8}$$

```
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```

```
>
_F1 := '_F1':

eqk := subs(lambda = ev(k), eq1);

solF := rhs(dsolve(eqk, _F1(x)));
```

$$eqk := \frac{d^2}{dx^2}\,\_F1(x) + \frac{\pi^2 k^2\,\_F1(x)}{L^2} = 0$$

$$solF := c_1 \sin\left(\frac{\pi k x}{L}\right) + c_2 \cos\left(\frac{\pi k x}{L}\right) \tag{9}$$

```
>
solF := subs(c__2 = 0, solF);
solF := subs(c__1 = 1, solF);
```

$$solF := c_1 \sin\left(\frac{\pi k x}{L}\right)$$

$$solF := \sin\left(\frac{\pi k x}{L}\right) \tag{10}$$

```
>
normF := simplify(
      solF/sqrt(int(solF^2, x = 0..L)),
      radical,
      symbolic
     ) :

normF := simplify(normF);
```

$$normF := \frac{\sin\left(\frac{\pi k x}{L}\right)}{\sqrt{\begin{cases} 0 & k = 0 \\ \frac{L\,(2\,\pi\,k - \sin(2\,\pi\,k))}{4\,\pi\,k} & otherwise \end{cases}}} \tag{11}$$

```
>
ef := unapply(normF, (k, x)) :

ef(k, x);
```

$$\frac{\sin\left(\frac{\pi k x}{L}\right)}{\sqrt{\begin{cases} 0 & k=0 \\ \frac{L\,(2\,\pi\,k-\sin(2\,\pi\,k))}{4\,\pi\,k} & \mathit{otherwise} \end{cases}}} \tag{12}$$

```
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```
> eq2 := diff(_F2(t),t) + a^2*lambda*_F2(t) = 0;
```

$$eq2 := \frac{d}{dt}\,\_F2(t) + a^2\,\lambda\,\_F2(t) = 0 \tag{13}$$

```
> eqt := subs(lambda = ev(k), eq2);
```

$$eqt := \frac{d}{dt}\,\_F2(t) + \frac{a^2\,\pi^2\,k^2\,\_F2(t)}{L^2} = 0 \tag{14}$$

```
> solT := rhs(dsolve(eqt, _F2(t)));
```

$$solT := c_1\,e^{-\frac{a^2\,\pi^2\,k^2\,t}{L^2}} \tag{15}$$

```
> solT := subs(_C1 = 1, solT);
```

$$solT := e^{-\frac{a^2\,\pi^2\,k^2\,t}{L^2}} \tag{16}$$

```
> tf := unapply(solT, (k,t)) :
tf(k,t);
```

$$e^{-\frac{a^2\,\pi^2\,k^2\,t}{L^2}} \tag{17}$$

```
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```

```
U := (x,t)    Sum(C(k)*tf(k,t)*ef(k,x),k=1..infinity);
```

$$U := (x,t) \quad \sum_{k=1}^{\infty} C(k) \cdot tf(k,t) \cdot ef(k,x) \tag{18}$$

```
> U(x,t) :
u(x,t) := U(x,t) :

u(x,t);
```

$$\sum_{k=1}^{\infty} \frac{C(k)\, \mathrm{e}^{-\frac{a^2 \pi^2 k^2 t}{L^2}} \sin\left(\frac{\pi k x}{L}\right)}{\sqrt{\begin{cases} 0 & k = 0 \\ \frac{L\,(2\pi k - \sin(2\pi k))}{4\pi k} & otherwise \end{cases}}} \tag{19}$$

```
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```
>
Ck := Int(phi(x)*ef(k,x),x=0..L) :
assume(L>0) :
Ck := value(Ck) :
Ck := simplify(Ck) :
Ck := subs(sin(Pi*k)=0,cos(Pi*k)=(-1)^k,Ck) :
Ck := simplify(Ck);
```

$$Ck := \frac{4\, c\, \sqrt{L\sim}\, \left(-1 + (-1)^k\right) \left(\begin{cases} \frac{1}{(-k)^{5/2} \sqrt{-2\pi k + \sin(2\pi k)}} & k < 0 \\ 0 & k = 0 \\ -\frac{1}{k^{5/2} \sqrt{2\pi k - \sin(2\pi k)}} & 0 < k \end{cases} \right)}{\pi^{5/2}} \tag{20}$$

```
>
C := unapply(Ck,k);
```

$$C := k \to \frac{4 \cdot c \cdot \sqrt{L\sim} \cdot \left(-1 + (-1)^{k}\right) \cdot \left(\begin{cases} \dfrac{1}{(-k)^{5/2} \cdot \sqrt{-2 \cdot k \cdot \pi + \sin(2 \cdot k \cdot \pi)}} & k < 0 \\ 0 & k = 0 \\ -\dfrac{1}{k^{5/2} \cdot \sqrt{2 \cdot k \cdot \pi - \sin(2 \cdot k \cdot \pi)}} & 0 < k \end{cases} \right)}{\pi^{5/2}} \quad \mathbf{(21)}$$

```
> C(k);
```

$$\frac{4\, c \sqrt{L\sim}\, \left(-1 + (-1)^{k}\right) \left(\begin{cases} \dfrac{1}{(-k)^{5/2} \sqrt{-2\, \pi\, k + \sin(2\, \pi\, k)}} & k < 0 \\ 0 & k = 0 \\ -\dfrac{1}{k^{5/2} \sqrt{2\, \pi\, k - \sin(2\, \pi\, k)}} & 0 < k \end{cases} \right)}{\pi^{5/2}} \quad \mathbf{(22)}$$

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```
> Sodd := subs(k = 2*m + 1, op(1, U(x, t))) :
  assume(m, integer) :
  Sodd := simplify(Sodd) :
  Sodd := subs((-1)^(2*m + 1) = -1, Sodd) :
  Sodd := simplify(Sodd) :
  Sodd := factor(Sodd);
```

$$Sodd := \frac{8\, c\, \mathrm{e}^{-\frac{a^{2} \pi^{2} (2\, m\sim + 1)^{2} t}{L\sim^{2}}} \sin\left(\frac{\pi\, (2\, m\sim + 1)\, x}{L\sim}\right)}{(2\, m\sim + 1)^{3}\, \pi^{3}} \quad \mathbf{(23)}$$

```
> uodd := Sum(Sodd, m = 0 ..infinity) :
  uodd;
```

$$\sum_{m\sim = 0}^{\infty} \frac{8\, c\, \mathrm{e}^{-\frac{a^{2} \pi^{2} (2\, m\sim + 1)^{2} t}{L\sim^{2}}} \sin\left(\frac{\pi\, (2\, m\sim + 1)\, x}{L\sim}\right)}{(2\, m\sim + 1)^{3}\, \pi^{3}} \quad \mathbf{(24)}$$

```
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```
sol := subs({a='a', L='L', m='m'}, uodd) :
sol;
```

$$\sum_{m\sim=0}^{\infty} \frac{8\, c\, \mathrm{e}^{-\frac{a^2 \pi^2 (2\, m\sim + 1)^2 t}{L\sim^2}} \sin\left(\frac{\pi (2\, m\sim + 1)\, x}{L\sim}\right)}{(2\, m\sim + 1)^3 \pi^3} \qquad \textbf{(25)}$$

>

>